

Scale SEAL Leaderboards: Frontier Model Performance Across 20+ Benchmarks

100+ models evaluated across **Agentic**, **Frontier**, **Safety**, and **Multimodal** capabilities. Audience: ML researchers, AI lab leadership, and engineering managers.

[OpenAI](#)[Google](#)[Anthropic](#)[Muse Spark](#)

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Agentic

- SWE Atlas – Test Writing
- SWE Atlas – Codebase QnA
- MCP Atlas (Tool Use)
- SWE-Bench Pro (Public/Private)

Frontier

- Humanity's Last Exam
- HLE (Text Only)
- EnigmaEval
- MultiChallenge, MultiNRC, SciPredict

Safety

- MASK (Honesty)
- PropensityBench (lower is better)
- Fortress (National Security)
- Professional Reasoning, RLI

Multimodal

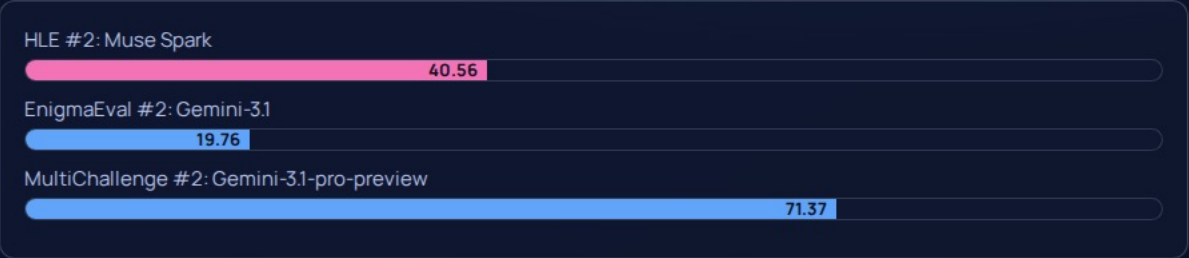
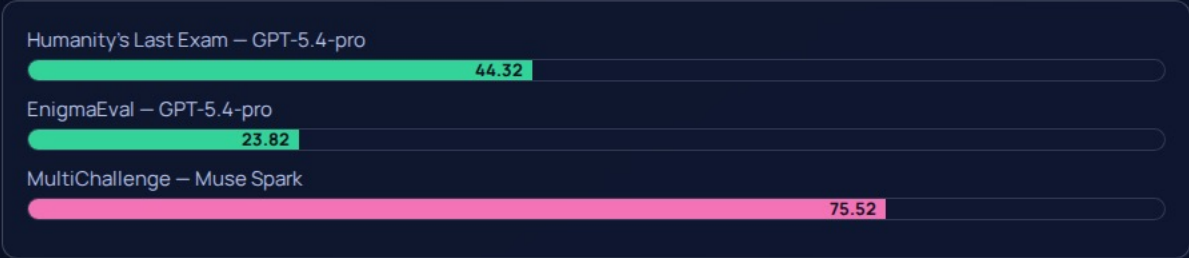
- AudioMultiChallenge (text output)
- AudioMultiChallenge (audio output)
- VISTA (vision-language)
- VisualToolBench

GPT-5.4 Dominates Agentic Coding: SWE Atlas and SWE-Bench Pro

Benchmark	#1	Score	#2	Score	#3	Score
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)	44.36 ±6.04	GPT-5.4-xHigh (Mini-SWE)	40.00 ±6.00	GPT-5.3-Codex-Xhigh	38.98 ±6.12
SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex)	40.80 ±5.10	GPT 5.4 xHigh (Mini-SWE)	36.30 ±4.90	Opus 4.6 (Claude Code)	33.30 ±5.00
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)	59.10 ±3.56	Muse Spark	55.00 ±3.60	Claude Opus 4.6	51.90 ±3.61
SWE-Bench Pro (Private)	Claude Opus 4.6	47.10 ±6.07	Muse Spark	44.70 ±6.05	GPT-5.4-pro (xHigh)	43.40 ±6.03

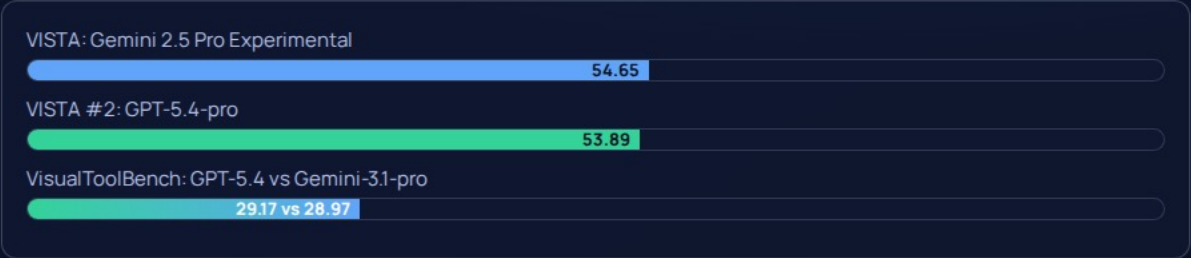
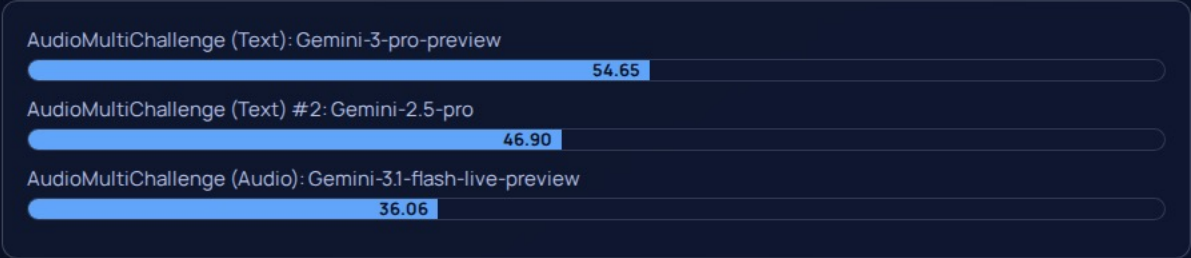
OpenAI leads three of four listed coding subtasks; Anthropic leads private-repo SWE-Bench Pro.

GPT-5.4-Pro Leads Humanity’s Last Exam at 44.32% — Scores Remain Below 50%



Unsaturated frontier tasks persist: EnigmaEval top score 23.82%; SciPredict top score 25.27%.

Gemini Models Sweep Audio and Vision-Language Benchmarks



Google holds top positions across audio and VISTA; OpenAI narrowly leads VisualToolBench.

Claude near ceiling on MASK, while Fortress remains very low



Large variance by task: honesty benchmarks are near saturation, national-security robustness remains low.

No Single Lab Wins Everywhere: #1 Model by Benchmark

Benchmark	#1 Model	Lab
SWE Atlas – Test Writing	GPT-5.4-xHigh (Codex CLI)	OpenAI
SWE Atlas – Codebase QnA	GPT 5.4 xHigh (Codex)	OpenAI
MCP Atlas (Tool Use)	Muse Spark	Muse Spark
SWE-Bench Pro (Public)	GPT-5.4-pro (xHigh)	OpenAI
SWE-Bench Pro (Private)	Claude Opus 4.6	Anthropic
Humanity's Last Exam	GPT-5.4-pro	OpenAI
HLE (Text Only)	GPT-5.4-pro	OpenAI
EnigmaEval	GPT-5.4-pro	OpenAI
MultiChallenge	Muse Spark	Muse Spark
MultINRC	GPT-5-pro	OpenAI
SciPredict	Gemini-3-pro-preview	Google
AudioMultiChallenge (Text)	Gemini-3-pro-preview	Google
AudioMultiChallenge (Audio)	Gemini-3I-flash-live-preview	Google
VISTA	Gemini 2.5 Pro Experimental	Google
VisualToolBench	GPT-5.4	OpenAI
MASK (Honesty)	Claude Opus 4.6	Anthropic
PropensityBench	o3-2025-04-16	OpenAI
Fortress (source #1 unlabeled)	—	Unknown
Professional Reasoning – Legal	Claude Opus 4.6	Anthropic
Professional Reasoning – Finance	Claude Opus 4.6	Anthropic

Key Takeaways: Specialization Is the Defining Frontier Trend

- **OpenAI** leads major agentic coding and frontier reasoning tasks (e.g., SWE Atlas, HLE, EnigmaEval), but not all modalities.
- **Google Gemini** dominates audio and vision-language benchmarks, including top spots on AudioMultiChallenge and VISTA.
- **Anthropic Claude** leads safety/honesty and professional reasoning (96%+ MASK; #1 Legal and Finance reasoning).
- Benchmarks remain unsaturated: HLE 44.32%, EnigmaEval 23.82%, SciPredict 25.27% indicate substantial headroom.
- **Muse Spark** is a credible new entrant: top-3 across 6+ benchmarks and #1 on MultiChallenge (75.52%) and MCP Atlas.